2025 Jan Oppe 1 - Set 2

Section 1 - problem 1

Check if a number divides two other numbers

PROBLEM STATEMENT

Write a function **is_both_divisible_by** that takes three integers a, b, and c as input and checks if a and b numbers are divisible by c.

Assume that c number is non-zero.
The function should return a boolean value.

EXAMPLE & EXPLANATION

Given the inputs a = 10, b = 15, and c = 5:
Explanation:

- The number 10 is divisible by 5 (10 % 5 = 0).
- The number 15 is divisible by 5 (15 % 5 = 0).

Since both are divisible, the result is True.
So, **is_both_divisible_by**(10, 15, 5) should return True.

Given the inputs a = 10, b = 13, and c = 5:
Explanation:

- 10 is divisible by 5.
- 13 is NOT divisible by 5 (13 % 5 = 3).

Since one is not divisible, the result is False.
So, **is_both_divisible_by**(10, 13, 5) should return False.

NOTE: This is a function type question, you don't have to take input or print the output, just have to complete the required function definition.

THE PROBLEM SPECIFICATION

INPUT

```
a (int)
b (int)
c (int)
```

Three integer arguments.
Constraint: c is non-zero.

→

THE BLACK BOX

Check if both a and b are divisible by c.

→

OUTPUT

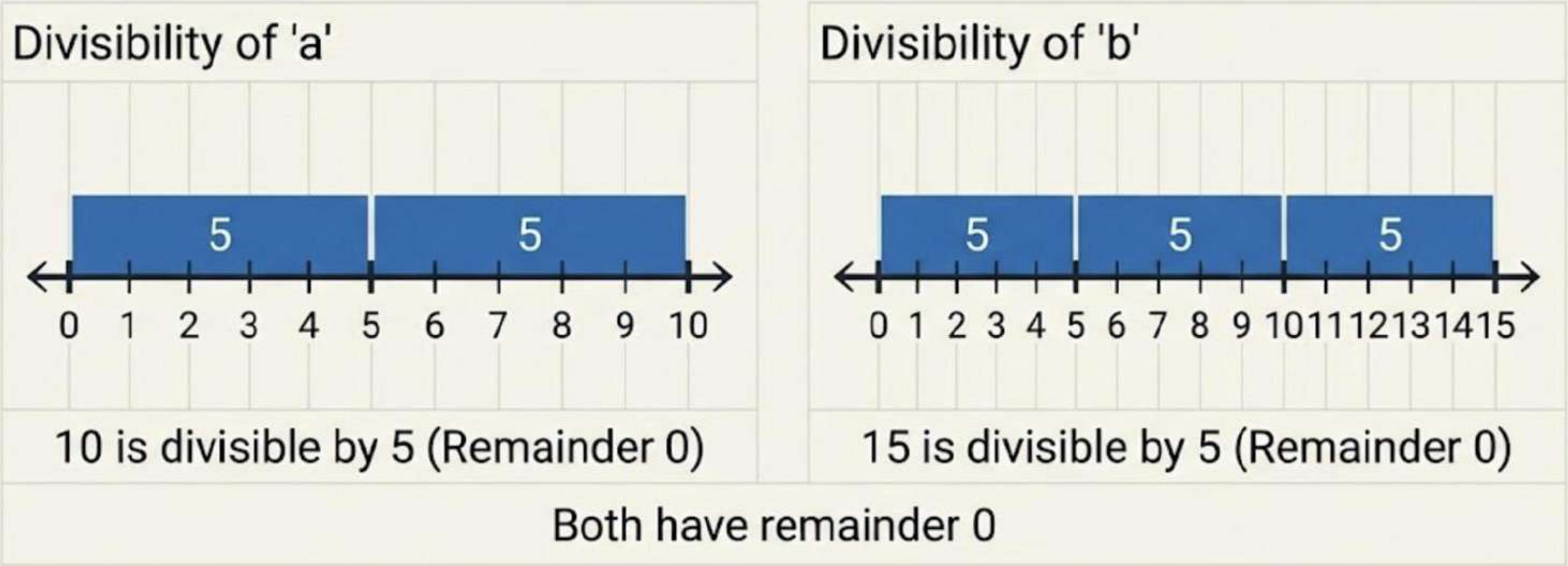
```
True or False
```

Single Boolean Return Value.

Note: This is a function type question, complete the required function definition.

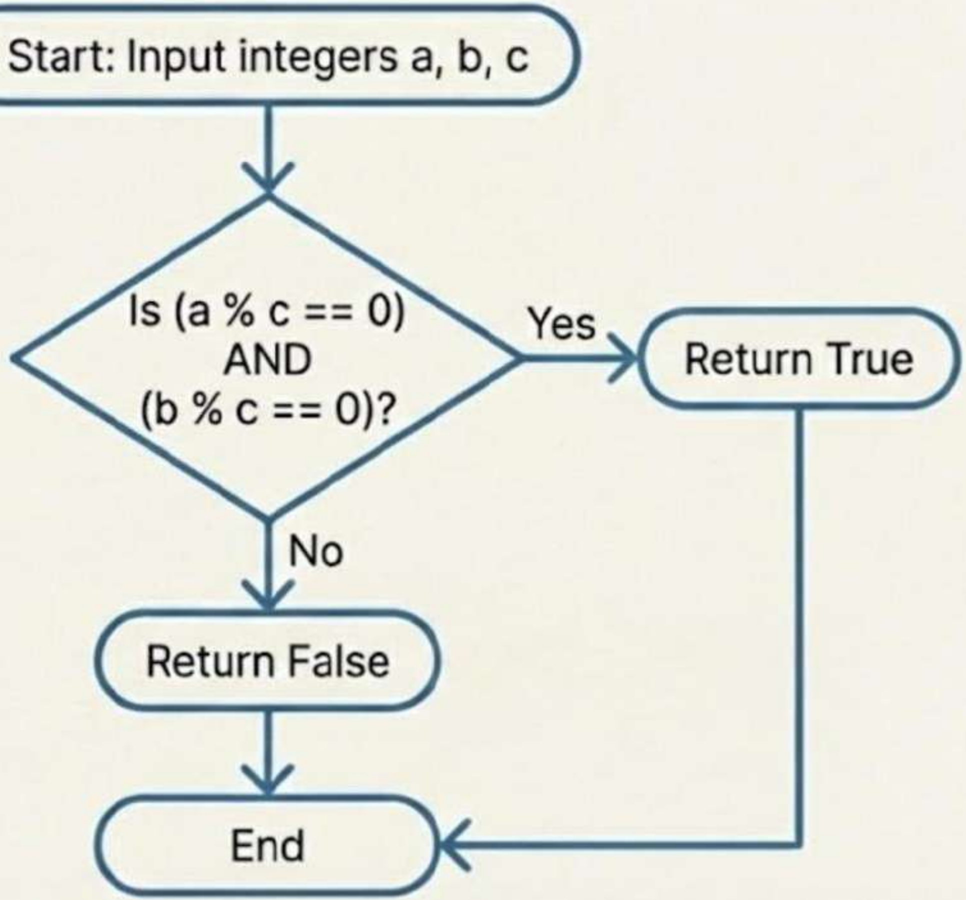
Visualizing Divisibility

Example Data: a = 10, b = 15, c = 5



Result (True, True) → Overall: True

The Logic Flowchart



The Explanatory Approach

Step-by-Step implementation for clarity.

```python
def is_both_divisible_by(a, b, c):
    # Step 1: Check divisibility for the first number 'a'
    a_is_divisible = (a % c == 0)

    # Step 2: Check divisibility for the second number 'b'
    b_is_divisible = (b % c == 0)

    # Step 3: Combine results to check if *both* are divisible
    both_divisible = a_is_divisible and b_is_divisible

    return both_divisible
```

<-- Modulo operator gives remainder. True if remainder is 0.

Intermediate variables improve debugging.

Logical AND operator ensures both conditions must be True.

Explicit return of the final boolean result.

The Pythonic Optimization

Leveraging built-ins for concise logic.

Type Hinting for clarity.

Modulo Operator & Boolean Logic.

```
def is_both_divisible_by(a: int, b: int, c: int) -> bool:
    return a % c == 0 and b % c == 0
```

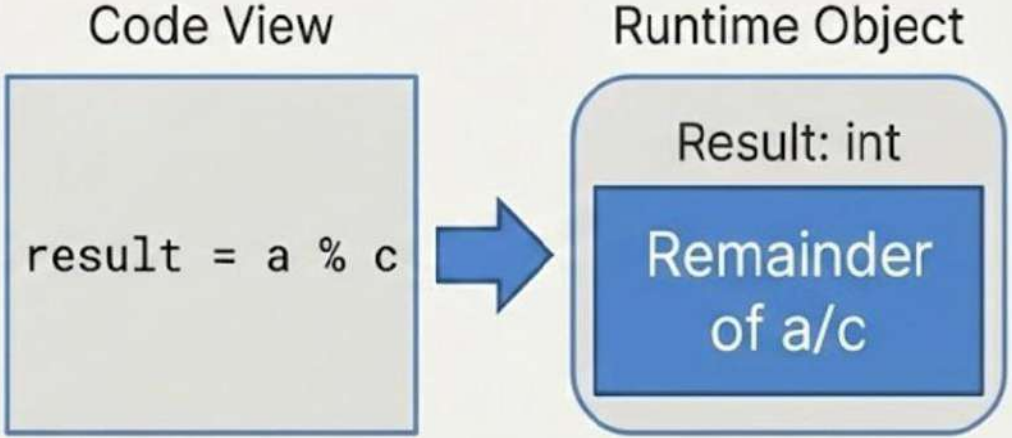
Single-line execution for direct boolean check.

Result: Identical output, cleaner syntax

Concept Spotlight: Modulo Operator (%)

Understanding the remainder of division.

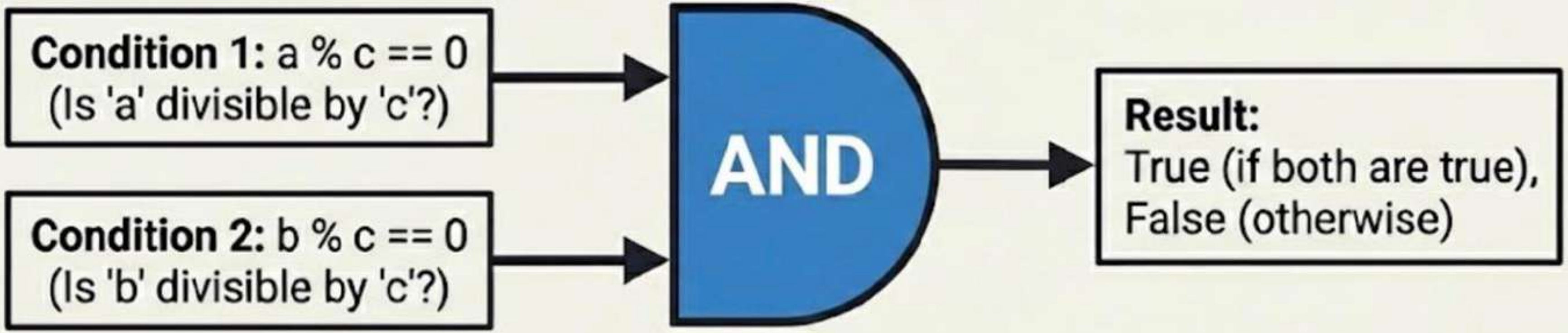
In Python, the modulo operator % returns the remainder of a division. For example, `10 % 5` results in 0 because 5 divides 10 perfectly with no remainder. It is essential for checking divisibility.



Application: (10 % 5) -> 0 (Since 5 divides 10 perfectly)

Concept Spotlight: The Power of Logical AND

Why both conditions must be true.



Scenario A:
is_both_divisible_by(10, 15, 5)
10%5==0 (T) & 15%5==0 (T)
Result: T AND T = True

Scenario B:
is_both_divisible_by(10, 13, 5)
|10%5==0 (T) & 13%5!=0 (F)
Result: T AND F = False

Scenario C:
is_both_divisible_by(19, 15, 5)
|19%5==0 (F) & 15%5==0 (T)
Result: F AND T = False

Robustness Rule: Use the logical ‘and’ operator to check that both divisibility conditions are met. **`return (a % c == 0) and (b % c == 0)`**.

Solution Comparison

Step-by-Step Approach

- High Readability.
- Clear intermediate steps.
- Best for: Beginners & Debugging.

```
is_a_divisible = (a % c == 0)
is_b_divisible = (b % c == 0)
return is_a_divisible and is_b_divisible
```

Concise One-Liner

- Mathematical purity.
- Fewer memory assignments.
- Best for: Production & Competitive Coding.

```
return (a % c == 0) and (b % c == 0)
```

The logical `and` operator returns `True` only if both conditions (a is divisible by c AND b is divisible by c) are `True`.

Key Takeaways

01

Problem: Divisibility Check

Write a function `is_both_divisible_by(a, b, c)` to check if both integers 'a' and 'b' are divisible by non-zero integer 'c'.

02

Core Logic: Modulo Operator

Use the modulo operator (%) to check for divisibility. `a % c == 0` means 'a' is divisible by 'c'. Both must be true.

03

Return Value: Boolean

The function must return `**True**` only if both conditions are met, otherwise `**False**`. Direct boolean calculation is efficient.

Final Syntax Cheat Sheet

```
return (a % c == 0) and (b % c == 0)
```